

LECTURE 2: GANs

SHARP IMAGES, UNSTABLE TRAINING

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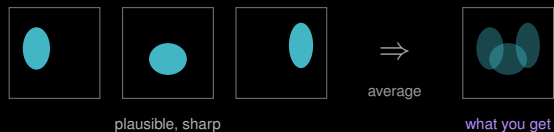
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The GAN: sharp, but unstable

WHY THE VAE BLURS

A GAN samples *exactly* like the VAE of Lecture 1: draw z , run one network. Only the **training signal** differs, and that is what fixes the blur.

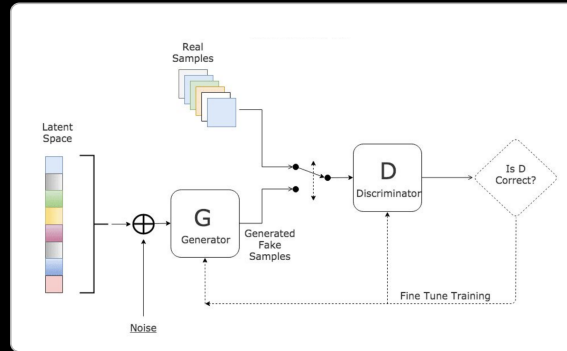
Given a latent z , many images are plausible. Squared error does not pick one — it returns their **average**.



So stop averaging

Replace the hand-written pixel loss with a **learned** one: train a second network to tell real from generated, and make the generator's job to fool it. A blurry sample is trivially caught, so blur stops being an option.

GAN: GENERATOR VS. DISCRIMINATOR



Generator

Maps noise z to an image.
Never sees real data — its only signal is D 's verdict.

Discriminator

A binary classifier, real vs. fake. Exactly the model from Module 2, used as a loss function.

The point

The loss is **learned and moving**. As G improves, D must find subtler tells.

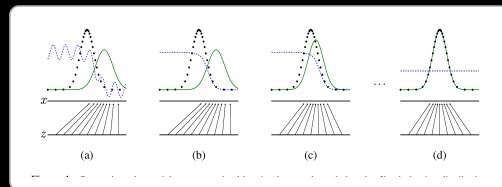
THE ADVERSARIAL GAME

Minimax objective

$$\min_G \max_D \mathbb{E}[\log D(x)] \\ + \mathbb{E}[\log (1 - D(G(z)))]$$

D maximises (catch the fakes); G minimises (fool D).

```
for x in loader:
    f = G(sample_z())
    d_re = bce(D(x), 1)
    d_fa = bce(D(f.detach()), 0)
    step(d_re + d_fa, opt_d)
    step(bce(D(f), 1), opt_g)
```



Black dotted = p_{data} , green = p_g , blue dashed = D .
From (a) to (d), p_g closes on p_{data} until $D \equiv \frac{1}{2}$ — the discriminator can do no better than a coin flip.

WHY GANS ARE HARD TO TRAIN

Mode collapse

G discovers one output that reliably fools D and emits it for every z . Samples look great and are all the same — disastrous when the modes you dropped are the rare pathologies.

Vanishing gradient

If D wins too early it returns 0 with confidence, and G receives almost no gradient. The two networks must improve *together*.

No loss curve to read

In every other lecture, falling loss meant progress. Here the loss is a **game score** between two moving players — it can sit flat while samples improve, or fall while they collapse.

You evaluate by **looking at samples**, which is exactly the problem the evaluation section has to solve.

Mitigations: balanced learning rates, label smoothing, spectral norm, WGAN-GP.